

- From next week: likely shift of Wed/Thu class to Fri. Stay tuned
 - Email me (upendra@cmi) any time
 - Tutorial groups and 1st HW soon
- [All problems stated in class and those in Artin are expected!!]

We asked: Linear algebra is about? First: solving $Ax=b$

Then: vectors and matrices

Today: More language about vectors and matrices, through analysis of solving $Ax=b$

words to come : { more examples of vector spaces: subspace of $\mathbb{R}^n$
linear combination, span, linear(in)dependence
basis, dimension

Review Elementary row operations on any matrix M

Let $R_i = i^{\text{th}}$ row, $R_j = j^{\text{th}}$ row

① $R_i \leftrightarrow R_j$	② $c R_i \quad c \neq 0$	③ $R_i + a R_j$
Switch two rows	Scale i^{th} row	Replace R_i by $R_i + a R_j$

Answer to yesterday's exercise/observation for later ($M = r \times s$)

The result of any of ①, ②, ③ is the same as EM , where $E = \begin{cases} r \times r \text{ matrix obtained by doing the same operation (①, ② or ③)} \\ \text{to the } r \times r \text{ identity matrix!} \end{cases}$

Thus,

$E = \text{resp.},$

called Elementary matrices

①
$$\begin{matrix} & i & j \\ i & \begin{bmatrix} 1 & & \\ & 0 & 1 \\ & & \ddots \\ & 1 & 0 \\ & & & 1 \end{bmatrix} \\ j & \end{matrix}$$

②
$$\begin{matrix} & i & j \\ i & \begin{bmatrix} 1 & & \\ & c & \\ & & 1 \end{bmatrix} \\ j & \begin{bmatrix} 0 & & \\ & & 1 \end{bmatrix} \end{matrix}$$

③
$$\begin{matrix} & j \\ i & \begin{bmatrix} 1 & & \\ & \ddots & \\ & & a \\ & 0 & 1 \end{bmatrix} \\ j & \begin{bmatrix} 0 & & \\ & & 1 \end{bmatrix} \end{matrix}$$

- Row operations are reversible!
- Each matrix can be transformed by row opⁿ to one in (reduced) echelon form as follows

(type ① operation)

- (i) pick leftmost nonzero column. Switch rows_k to get a nonzero entry in top row. Scale to make this entry 1 (type ②)
- (ii) Kill entries below this 1 (type ③ operation)
- (iii) set aside the first row and repeat steps (i) and (ii) till only zeros are left in any remaining bottom rows

This gives an echelon form. Leading 1 in any nonzero row is called a pivot

- (iv) To get reduced echelon form kill all entries above each pivot again by type ③.

($A = r \times s$ matrix) Solving $Ax = b$ using EF or REF of $[A|b]$

Ex. 3 eqⁿ in 4 variables i.e. $A = 3 \times 4$ matrix

Suppose

$$\left[A_{3 \times 4} \begin{vmatrix} b_1 \\ b_2 \\ b_3 \end{vmatrix} \right] \xrightarrow[\text{op}^n]{\text{row}} \left[\begin{array}{cccc|c} 1 & p & 0 & q & c_1 \\ 0 & 0 & 1 & r & c_2 \\ 0 & 0 & 0 & 0 & c_3 \end{array} \right] \quad \begin{array}{l} \text{Each } c_i \text{ is of form} \\ \alpha b_1 + \beta b_2 + \gamma b_3 \text{ for} \\ \text{some scalars } \alpha, \beta, \gamma \end{array}$$

$x_1 \quad x_2 \quad x_3 \quad x_4$

Solutions to both systems are the same

- Either system is consistent ($\Leftrightarrow$) $c_3 = 0$
- Suppose $c_3 = 0$. Then solution set is as follows

free variables: x_2, x_4

pivot variables: x_1, x_3

}

$(-px_2 - qx_4 + c_1, x_2, -rx_4 + c_2, x_4)$

where x_2, x_4 are arbitrary

General solution

$$\begin{array}{l} x_1 = -p x_2 - q x_4 + c_1 \\ x_2 = x_2 \\ x_3 = -r x_4 + c_2 \\ x_4 = x_4 \end{array} \quad \left| \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} -p \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -q \\ 0 \\ -r \\ 1 \end{bmatrix} + \begin{bmatrix} c_1 \\ 0 \\ c_2 \\ 0 \end{bmatrix} \right.$$

In general

- $Ax=b$ inconsistent $(\Rightarrow)$ there is a pivot in the last column of $[A|b]$
- $Ax=b$ is consistent for all possible vectors in $\mathbb{R}^r$
 - $(\Rightarrow)$ any echelon form of A has no row consisting of all zeros
 - $(\Rightarrow)$ Every row of any echelon form of A has a pivot

Ex Number of pivots in any echelon form of A is the same

number of pivots in (echelon form of) $A \leq \min(r, c)$

Added later: Either equality ($\# \text{ pivots} = r$ or $= c$) is interesting, as we will see.

We can neatly separate discussion of consistency of $Ax=b$ and the dimension of its solution set (assuming consistency) by considering the associated homogeneous system $Ax=0 \rightarrow$ This system is always consistent! (namely $x=0$ is always a solution)

Key point Suppose v is a solution to $Ax=b$
(v is given s.t. $Av=b$)

Then Claim:

$$(\text{Solution set to } Ax=b) = (v + \text{solution set to } Ax=0)$$

$$\text{i.e. } \underbrace{\{x \mid Ax=b\}}_S = \underbrace{\{v+y \mid Ay=0\}}_T$$

$$Ax=b$$

$$Ax=0$$

e.g.

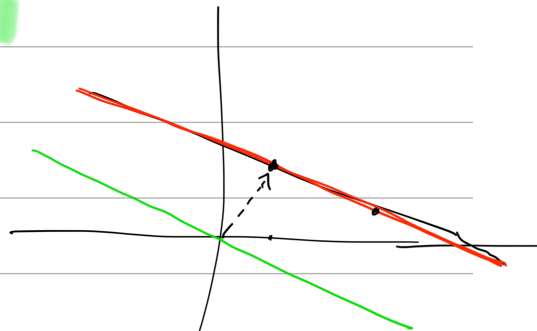
$$3x+2y=7$$

$\rightsquigarrow$

$$3x+2y=0$$

$$v = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \text{ specific/given solution}$$

$$\begin{aligned} &\{ \text{every point on green line} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} \} \\ &= \{ \text{every point on red line} \} \end{aligned}$$



Proof of claim $S=T$. We will show ⁽ⁱ⁾ $S \subset T$, ⁽ⁱⁱ⁾ $T \subset S$

(i) Let $x \in S$. We need to show $x \in T$ i.e.
 so we know that $Ax=b$ $\implies x = v + (\text{some } y) \text{ s.t. } Ay=0$
 We are forced to try $y := x - v$

Does it work? Need to check $Ay=0$

$$\text{Check: } A(x-v) = Ax - Av$$

$$\therefore S \subset T \qquad = b - b = 0 \quad \checkmark$$

(ii) Let $u \in T$. We know $u = v + y$ where $Ay=0$

Does $u \in S$? i.e. We need to check if $Au=b$

$$\underline{\text{Check}} \quad Au = A(v+y) = Av + Ay$$

$$= b + 0 = b \quad \checkmark$$